

The Epistemic Impossibility of Predictor-Centric Architectures

Tony Mason [Collaborator Name]

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Abstract

Large language model “hallucinations” are commonly framed as failures introduced by alignment techniques or insufficient training data. In this paper, we argue that this framing is incorrect. We present a formal impossibility result demonstrating that **epistemic honesty is unreachable** for the class of predictor-centric architectures (M_{closed}) aligned via standard reward modeling.

We decompose this failure into two theorems. **Theorem 1 (Representational Impossibility)** proves that a policy conditioning solely on the query cannot satisfy epistemic requirements that depend on external world states. **Theorem 2 (Learnability Impossibility)** proves that even if the architecture is expanded to include retrieval (Grounded Policy), a resource-bounded supervisor cannot distinguish truthful answers from plausible fabrications, rendering the honest policy unlearnable via preference optimization. We validate these theoretical results with forensic artifacts from frontier model interfaces (Grok, Gemini, OLMo), identifying a “**Deadlock of Veracity**” where human operators are architecturally forced to reinforce fabrication.

1 System Definitions

Let $\mathcal{Q}$ be the space of queries, $\mathcal{R}$ be the space of responses (including $\perp$ for abstention), and $\mathcal{W}$ be the set of possible world states.

Definition 1.1 (Predictor-Centric Policy). For any query $q \in \mathcal{Q}$, $r \in \mathcal{R}$, and world state $w \in \mathcal{W}$, a predictor-centric model is defined by a stochastic policy $\pi : \mathcal{Q} \rightarrow \Delta(\mathcal{R})$, where $\pi(r|q)$ denotes the probability of generating response r given query q . Crucially, π does not take w as an input at inference time.

Definition 1.2 (Truth Status). For any query q and world state $w \in \mathcal{W}$, we define a binary truth status $U(q, w)$:

- $U(q, w) = 1$ (**Determined**): The query has a verifiable answer in w .
- $U(q, w) = 0$ (**Underdetermined**): The query is unanswerable or the answer is unknown in w .

Definition 1.3 (Epistemic Honesty). A policy π is **Epistemically Honest** with margin ϵ relative to world w if:

1. If $U(q, w) = 1$, there exists a correct response r_{corr} such that $\pi(r_{corr}|q) \geq 1 - \epsilon$.
2. If $U(q, w) = 0$, $\pi(\perp|q) \geq 1 - \epsilon$.

$A \neq B$

2 Theorem 1: Representational Impossibility

This theorem establishes that the predictor-centric architecture itself is insufficient, regardless of training quality.

Condition 2.1 (Ambiguity). There exists a query q and two world states $w_A, w_B \in \mathcal{W}$ such that:

1. $U(q, w_A) = 1$ (Answerable in w_A).
2. $U(q, w_B) = 0$ (Unanswerable in w_B).

Theorem 2.1. *For any predictor-centric policy $\pi : \mathcal{Q} \rightarrow \Delta(\mathcal{R})$, it is impossible to satisfy Epistemic Honesty for both world states w_A and w_B simultaneously given the ambiguous query q , provided $\epsilon < 0.5$.*

Proof. 1. The policy $\pi(r|q)$ is a function of q alone. Thus, the distribution over responses is identical in both worlds: $\pi(\cdot|q, w_A) = \pi(\cdot|q, w_B) = \pi(\cdot|q)$.

2. To be honest in w_A , we require $\pi(r_{\text{corr}}|q) \geq 1 - \epsilon$.
3. To be honest in w_B , we require $\pi(\perp|q) \geq 1 - \epsilon$.
4. Since $r_{\text{corr}} \neq \perp$, these events are disjoint. The sum of probabilities would be $\geq 2(1 - \epsilon)$.
5. If $\epsilon < 0.5$, the sum exceeds 1, which violates the axioms of probability.
6. Therefore, no such policy π exists. □

Implication: A model that cannot “see” the world state (via retrieval or grounding) at inference time cannot be honest about it.

3 Theorem 2: Learnability Impossibility

This theorem establishes that even if the architecture is expanded to allow dependence on w (e.g., via RAG), the training protocol prevents learning the correct dependence if the supervisor is bounded.

Definition 3.1 (Grounded Policy). A grounded policy is a stochastic function $\pi : \mathcal{Q} \times \mathcal{D} \rightarrow \Delta(\mathcal{R})$, where $\mathcal{D}$ represents retrieved documents or observations of w available at inference time. This policy class *can* theoretically represent different behaviors for w_A and w_B .

Definition 3.2 (The Bounded Supervisor). The training process relies on a supervisor S that assigns reward R based on an observation function $Obs(q, r, w)$. The supervisor has a verification budget B .

$$Obs(q, r, w) = \begin{cases} (q, r, \text{Verify}(r, w)) & \text{if } C_{\text{verify}}(q, r, w) \leq B \\ (q, r, \emptyset) & \text{if } C_{\text{verify}}(q, r, w) > B \end{cases}$$

Condition 3.1 (The Hallucination Regime). There exists a plausible fabrication r_{fab} and contexts $(q, w_A), (q, w_B)$ from Condition 2.1 such that:

1. The cost to verify r_{fab} in both contexts exceeds B (e.g., obscure or non-existent citation).

2. Consequently, the supervisor observation is identical:

$$\text{Obs}(q, r_{fab}, w_A) = \text{Obs}(q, r_{fab}, w_B) = (q, r_{fab}, \emptyset)$$

Theorem 3.1. *No learning algorithm $\mathcal{A}$ optimizing reward R from a bounded supervisor S can converge to a Grounded Policy that satisfies Epistemic Honesty across both contexts w_A and w_B for the response r_{fab} .*

- Proof.*
1. The learning update $\Delta\theta$ is a function of the observation and reward.
 2. From Condition 2.2, the observations for r_{fab} are identical in both worlds.
 3. Therefore, the reward distribution and expected gradient updates are identical: $\mathbb{E}[\Delta\theta|w_A] = \mathbb{E}[\Delta\theta|w_B]$.
 4. The optimizer receives no signal to distinguish the cases.
 5. To satisfy Epistemic Honesty, the policy must converge to output r_{fab} in w_A and $\perp$ in w_B .
 6. However, since the gradient signal is identical, the policy must converge to the same behavior in both cases (either rewarding r_{fab} in both, or penalizing it in both).
 7. It is therefore impossible to learn the split behavior required for honesty.

□

Remark 3.1 (Empirical Bias). While the theorem proves only that the behavior must be *identical*, empirical observation shows a strong bias. Supervisors typically prefer “plausible completion” over “abstention” when truth is unknown. This effectively breaks the tie in favor of fabrication.

4 Empirical Diagnostics (Forensic Exhibits)

4.1 The Sycophancy Trap (Exhibit A: Grok)

Artifact: Forced choice between two lies regarding a non-existent Adam Smith paper.

The Failure: The RLHF interface forced the human operator to select a “preferred” lie. This validates **Theorem 2:** The supervisor (human) is bounded and the interface (protocol) cannot express the necessary distinction, forcing a gradient update that reinforces fabrication.

4.2 The Signal-to-Noise Asymmetry (Exhibit B: OLMo)

Artifact: 92% fabrication rate on fictitious papers.

The Failure: Base models produce coherent 1500-token fabrications with negligible “uncertainty” signaling. This validates **Theorem 1:** The predictor-centric base model, lacking access to world-state w , generates the statistically most probable continuation for the query context.

5 Architectural Implications

To escape these impossibility results, systems must adopt specific design principles:

Principle 5.1 (State Exteriority). The representation of validity must be separated from the representation of generation. The policy must explicitly condition on retrieved world-state w , not just query q .

Principle 5.2 (Verification Independence). Verification signals must originate from a channel orthogonal to the generation signal, capable of overriding the bounded supervisor’s preference for plausibility.

Principle 5.3 (Provenance Binding). Every output assertion must be structurally bound to a verifiable source.